

Apache OpenOffice

Version 4.1

Draw Guide

AOO Documentation Team

Chapter 4 **Changing Object Attributes**

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Acknowledgments

This book is updated from a previous version by the OOoAuthors team. The AOO Documentation Team appreciates the many contributions of Michael Scarborough and Michele Petrovsky.

Publication date and software version

Published December 20, 2025. Based on Apache OpenOffice 4.1.

Note for Mac Users

Some keystrokes and menu items differ on a Mac from those used in Windows and Linux. The table below gives some common substitutions for the instructions in this chapter. For a more detailed list, see the application Help.

<i>Windows/Linux</i>	<i>Mac equivalent</i>	<i>Effect</i>
Tools > Options menu selection	OpenOffice > Preferences	Access setup options
<i>Right-click</i>	<i>Control+click</i>	Open context menu
<i>Ctrl</i> (Control)	⌘ (Command)	Used with other keys
<i>F5</i>	<i>Shift+⌘+F5</i>	Open the Navigator
<i>F11</i>	<i>⌘+T</i>	Open Styles & Formatting window

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Toolbars and Menus

You can use the Properties deck of the Sidebar, the Line and Filling toolbar, or the context menu to change an object's attributes, such as color or border width.

Line and Filling Tools

The Line and Area panes of the Sidebar's Properties deck have tools for modifying the most important fill and line properties of an object. If the Sidebar is not visible, select **View > Sidebar**. You can then choose the Properties icon, which is the topmost of the four icons on the right edge of the Sidebar. The Line and Area panes will be visible when an object is selected, as we've shown in Figure 1.

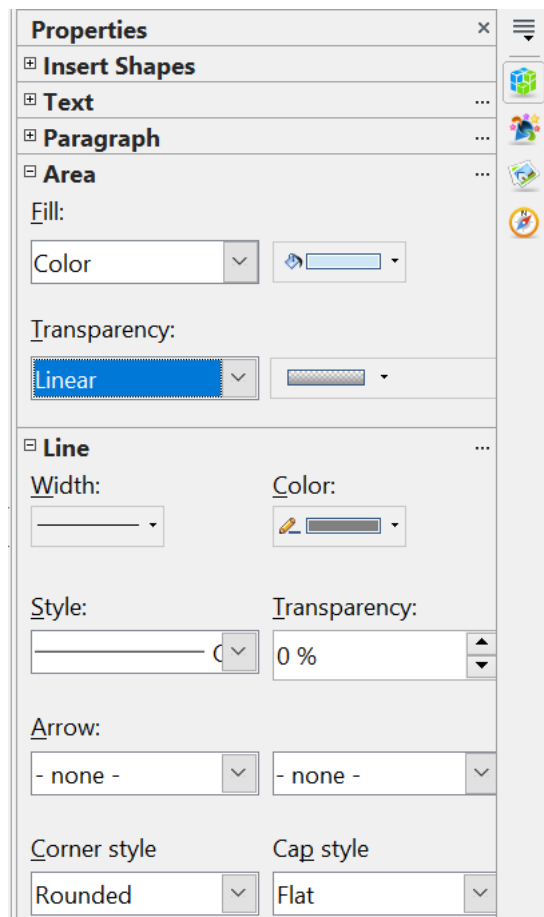


Figure 1: The Area and Line Panes of the Sidebar

The four boxes in the Area pane provide access to a large number of choices. The Fill type, which defaults to *Color*, can also have the following values: *None*, *Gradient*, *Hatching*, or *Bitmap*. Each of those choices changes the options in the adjacent drop-down list. Similarly, the Transparency list has eight options: *None*, *Solid*, *Linear*, *Axial*, *Radial*, *Ellipsoid*, *Quadratic*, and *Square*. For most of these, multiple parameters, such as the minimum and maximum transparency, are accessed through the adjacent drop-down list.

The tools on the Line pane are self-explanatory, except that the two boxes under the *Arrow* heading do not refer to the left and right ends of the line. They refer to the beginning and end of the line, which are determined when the line is drawn. If you draw the line from right to left, the beginning is on the right.

Most panes in the Sidebar show only the most commonly used tools. The full set of tools can be accessed by clicking the three dots at the top right of each pane, which brings up a dialog with all available settings. In particular, we will refer to the Area dialog and Line dialog several times.

The Line and Filling toolbar shown in Figure 2 provides many of the same tools as the Sidebar panes. If the Line and Filling toolbar is not visible, you can display it using **View > Toolbars > Line and Filling**. From here, you can edit the most common object attributes. You can also open the Line dialog by clicking on the Line  icon and the Area dialog by clicking on the Area  icon to see more options.

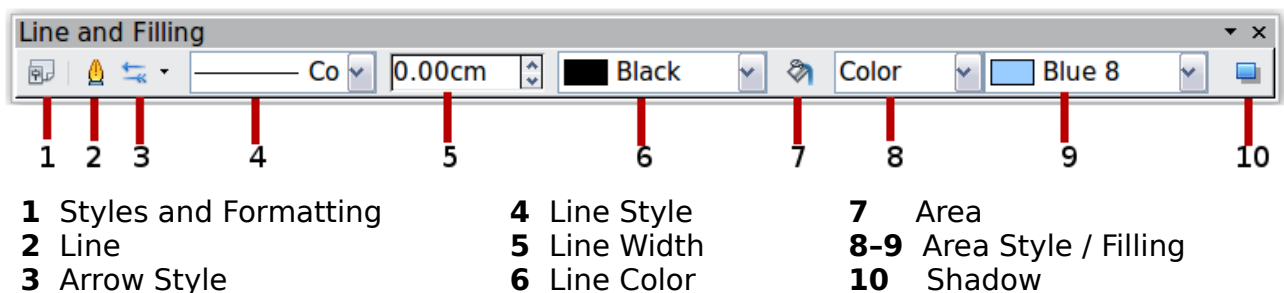


Figure 2: Line and Filling toolbar

Text Formatting Toolbar

When you select a block of text, the Line and Filling toolbar changes to show text formatting options like those in Figure 3. You can also display the Text Formatting toolbar by choosing **View > Toolbars > Text Formatting**.

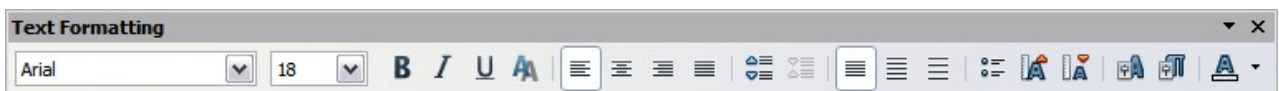


Figure 3: Text Formatting toolbar (when text is selected)

Since any object can contain text, the Properties pane of the Sidebar always provides the Text and Paragraph panes when an object is selected. These panes have most of the same tools as the Text Formatting Toolbar and, as usual, the complete set of tools can be accessed by clicking the three dots at the upper right of each pane.

The Context Menu

When an object is selected, you can right-click on the object to bring up a context menu (Figure 4). The context menu provides additional access to the options shown above and another way to change an object's attributes. The entries with a small arrow on the right-hand side contain a submenu.

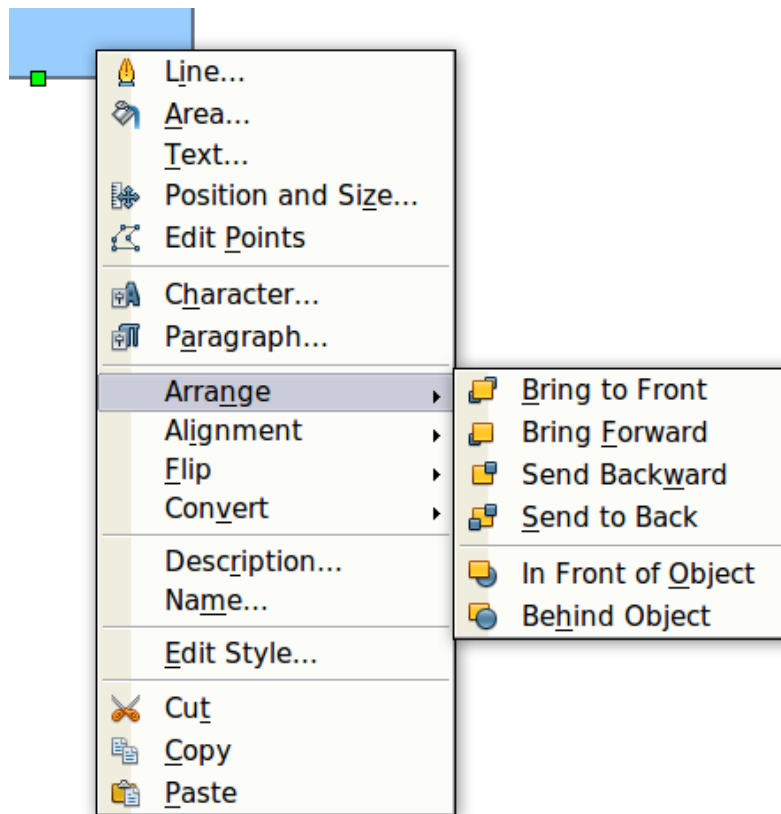


Figure 4: Right-click on an object to see the context menu

Editing Lines and Borders

Lines, arrows, and the borders of an object are all managed through the same dialog (Figure 6) except that the Shadow tab is only available if the selected object is a line. The properties on the other tabs apply to the borders of two-dimensional objects, but the shadow is a property of the Area of a two-dimensional object.

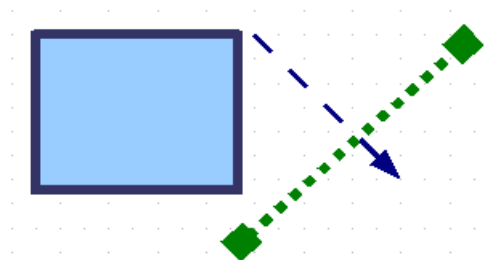


Figure 5: Lines and borders

You can change some properties from the Sidebar or from the Line and Filling toolbar. To see more options, click the three dots at the upper right of the Line pane of the Sidebar or right-click on the object and choose **Line** from the context menu. This opens the Line dialog.

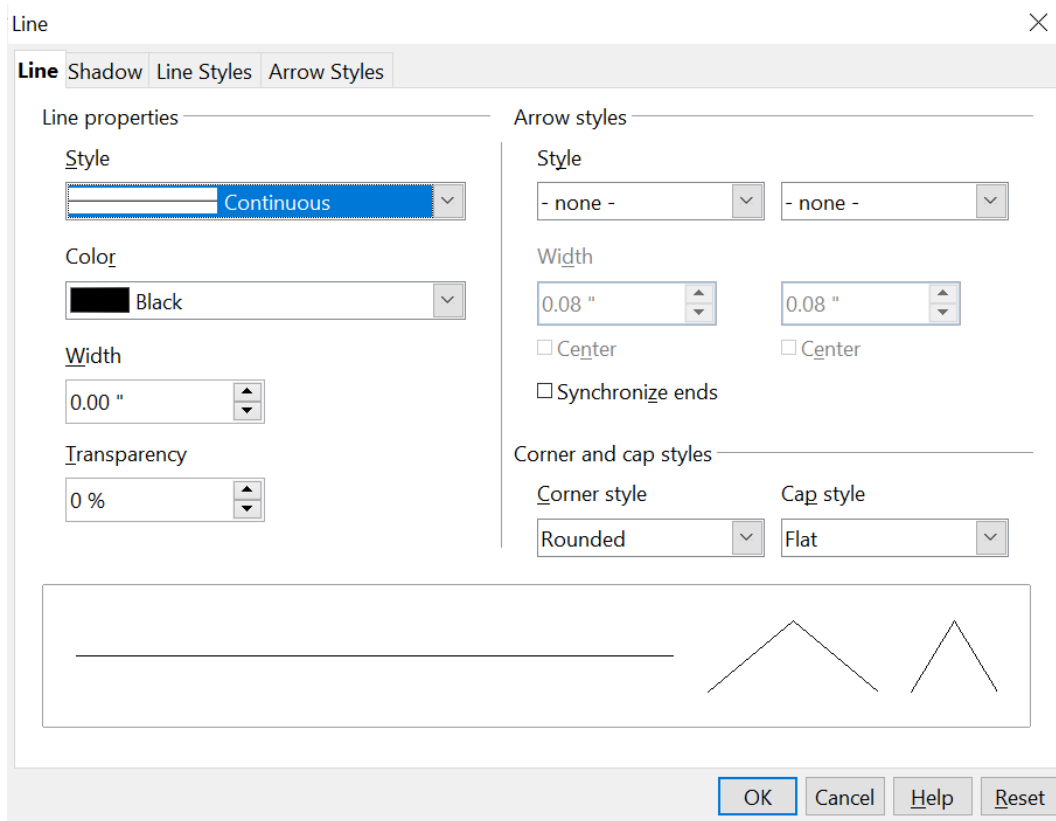


Figure 6: Line dialog .

Common Line Properties

In most cases, the property you want to change is the line's style (solid, dashed, invisible, and so on), its color, or its width, as is done in Figure 7. These options are all available from the Sidebar's Line pane (Figure 1) or the Line and Filling toolbar.

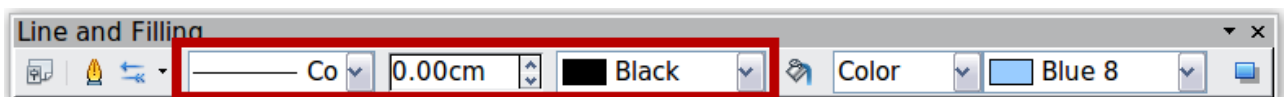


Figure 7: Common line options (style, width, color)

You can also edit these properties from the Line dialog. They are located on the first tab, in the left column (see Figure 6). From the Line dialog or the Sidebar, you can also change the line's transparency. Figure 8 illustrates different degrees of transparency.

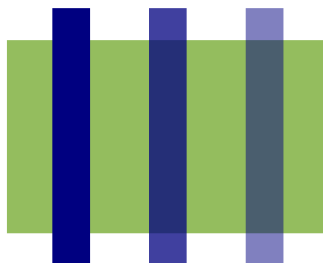
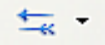


Figure 8: The vertical lines have different levels of transparency (0%, 25%, and 50%).

Drawing Arrows

Arrowheads and other line endings, usually referred to in this guide as arrows, are a line property. Select a line and click on one of the Arrow drop-down boxes in the

Sidebar or the **Arrow Style**  icon in the Line and Filling toolbar. This opens the Arrowheads menu shown in Figure 9.

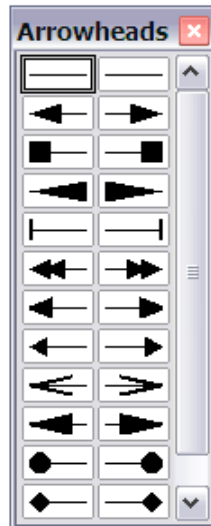


Figure 9: Arrowheads menu

There are several types of arrowheads available. Each end of a line can have a different arrowhead, or even no arrowhead. Remember that the selection on the left corresponds to the beginning of the line as it was drawn, not to the left end of the line.

Note Arrowheads are only applicable to lines. They have no effect on the border of an object.

In the Line dialog (Figure 6), *Arrow styles* on the right contain several options to fine-tune the arrow properties. If **Synchronize ends** is selected, both line endings will appear the same. The **Center** option, illustrated in Figure 10, places the center of the arrow over the endpoint of the line. If this option is not selected, the line ends at the far edge of the arrow. The following sketch illustrates the difference.

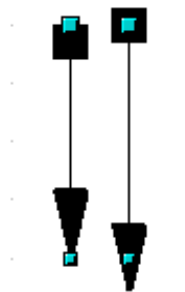


Figure 10: Default arrowheads (left) vs centered arrowheads (right)

Line Shadows

Lines can have a shadow property associated with them, as is done in Figure 11.

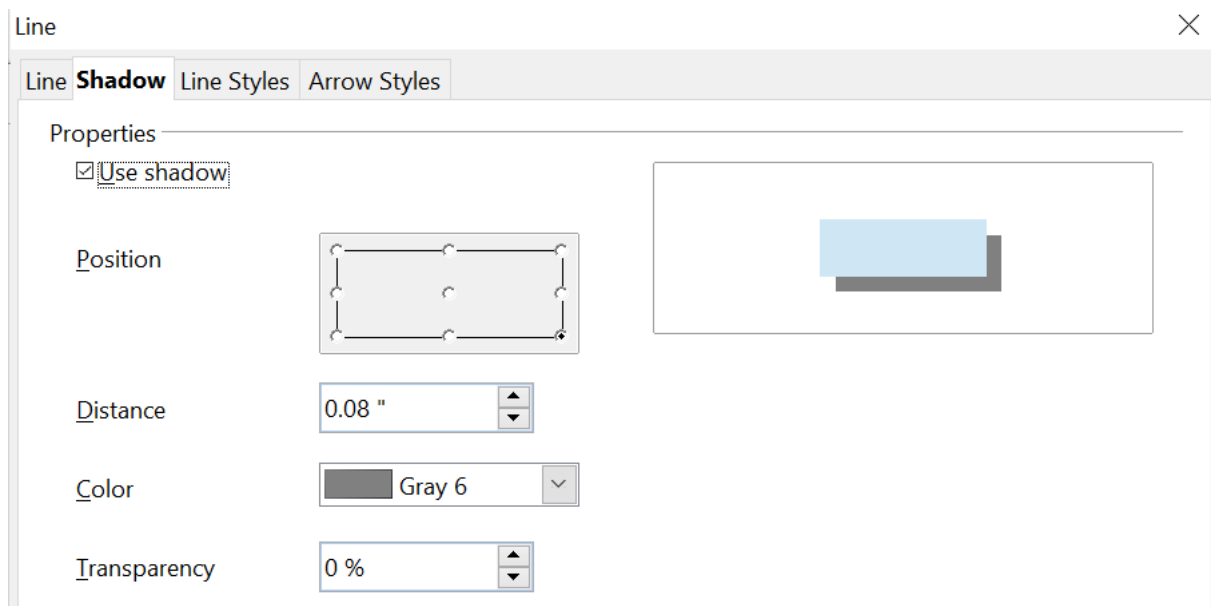


Figure 11: Shadow page of Line dialog

Check the *Use shadow* box to enable this feature. The properties shown—position, distance, color, and transparency—can be independently adjusted to your choice and the results seen in the preview window. You can also click on the Filling  icon on the **Line and Filling** toolbar—next to the area fill functions – to get a basic shadow.

Customizing Line and Arrow Styles

You are not required to use only the line and arrow styles provided by default in Draw. You can modify the styles and even create your own.

Customizing Line Styles

In the Line dialog, click on the **Line Styles** tab (see Figure 12). From here, you can customize the line styles or create your own by clicking on the **Add** button. You can change the length of the dashes, the space between them, and several other attributes.

Use the **Load Line Style**  and **Save Line Style**  icons to save a new definition or read one from disk (file extension `.sod`).

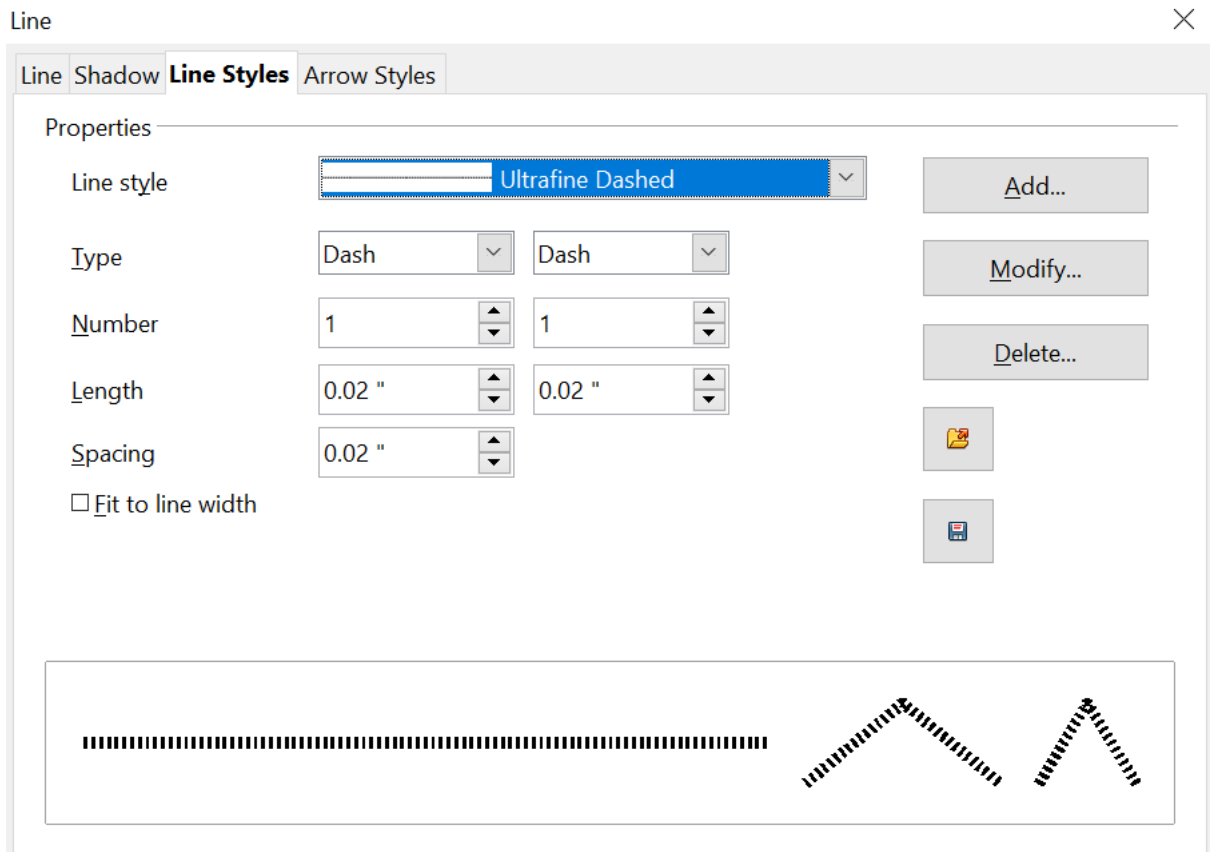
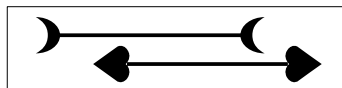


Figure 12: Editing line styles

Customizing Arrow Styles

You can also create your own arrowheads to provide some interesting effects, such as:



The first step is to draw a curve with the shape you want to use as the arrowhead, like we did in Figure 13.

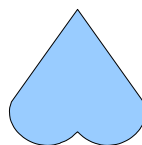


Figure 13: To create your own arrowhead, you must first draw a curve.

Note

The arrowhead must be a curve. A curve is something you could draw without lifting a pencil. For example, ☆ is a curve, but ☺ is not a curve. You can, however, draw forms that are not curves and then convert them to a curve.

Select the curve, open the Line dialog, and go to the **Arrow Styles** page. Click on **Add**, enter a name for the arrow style, and click **OK** (see Figure 14).

Tip

The part of the shape that should point in the direction of the line must be drawn facing upwards. In Figure 13, the top of the shape will point towards the “outside” of the line.

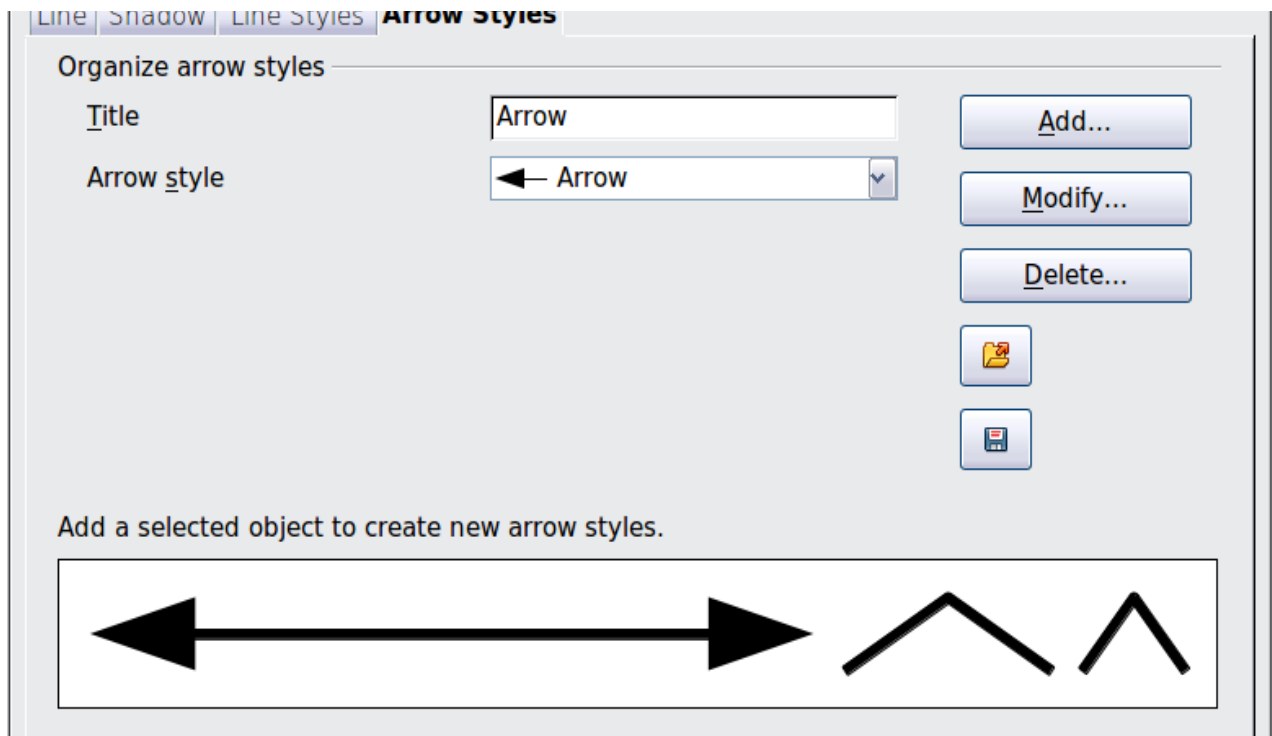


Figure 14: Adding an arrow style

Now you can access the new style from the *Arrow style* list (Figure 15) or the *Arrowheads* dialog (Figure 9).



Figure 15: Arrow style list

Editing the Inside (Fill) of an Object

The OpenOffice term for the inside of an object is **Area fill**. The area fill of an object can be a uniform color, a gradient, a hatching pattern, or an image. It can be made partly or wholly transparent and can throw a shadow.

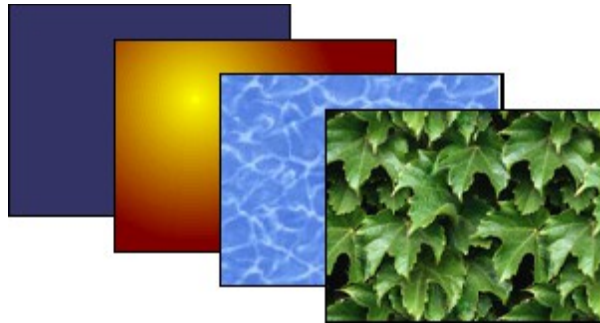


Figure 16: Different types of area fill

Common Fill Properties

In most cases, you will choose one of the standard fill options, whether it is a color, a gradient, or an image. These options are all available from the Area pane of the Sidebar or the Line and Filling toolbar.

If you want no fill at all, select the object you wish to edit and set the Fill to **None** on the Area pane of the Sidebar.

Fill with a uniform color

Select the object you wish to edit. On the Area pane, set the Fill to Color or on the Line and Filling toolbar, select **Color** from the pull-down list at the right of the paint can. In either case, choose a color from the right-hand menu, as we did in Figures 17 and 18.

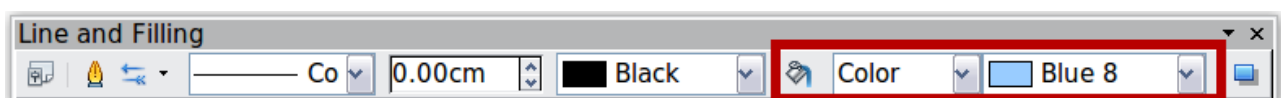
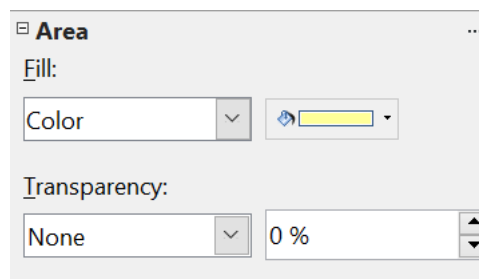


Figure 17: Common fill options in the Sidebar and Line and Filling toolbar

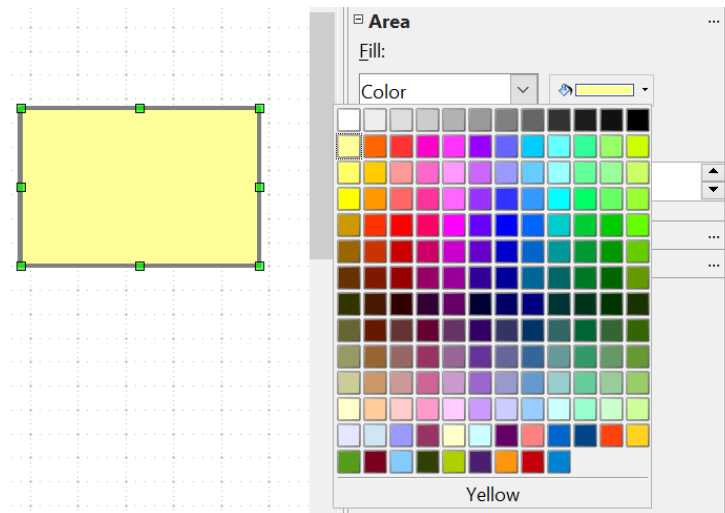


Figure 18: Filling with a color using the Sidebar

Fill with a Gradient

Select the object you wish to edit. Set the Fill type to **Gradient** and then choose a gradient from the right-hand menu, as Figure 19 illustrates.

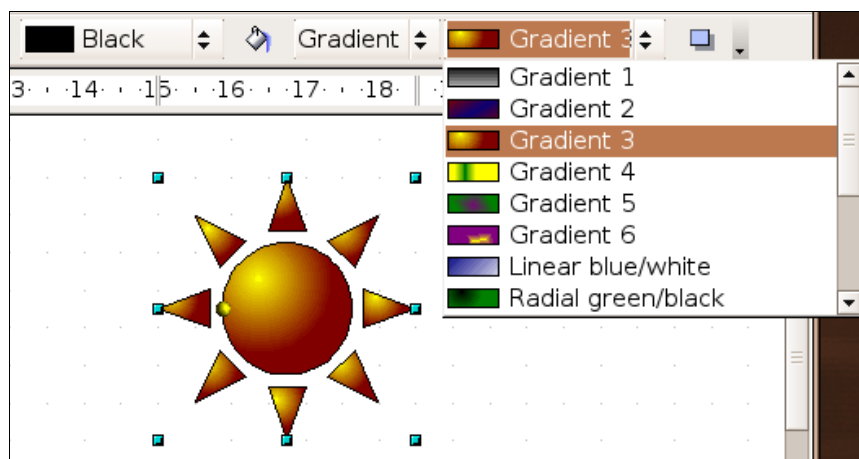


Figure 19: Filling with a gradient using the toolbar

Fill with a Line Pattern or an Image

Objects can also be filled with line patterns or an image. The OpenOffice term for line patterns is Hatching. Images used as fill patterns must be bitmaps, that is, images stored as a pattern of pixels rather than as vector images. Select the object you wish to edit. In the Sidebar or on the Line and Filling toolbar, set the Fill to **Hatching** or **Bitmap**. Then choose one of the available patterns from the neighboring drop-down list.

Adding a Shadow to an Area

To get the default shadow for an object, click on the Filling  icon on the **Line and Filling** toolbar—next to the area fill functions (Figure 20).

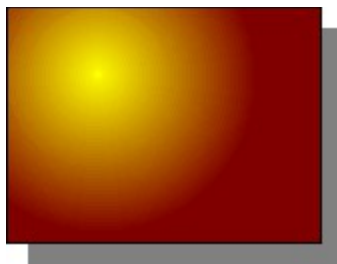


Figure 20: Adding a shadow

You can control the position, distance, color, and transparency of the shadow on the Shadow tab of the Area dialog. This is identical to the shadow dialog in Figure 11.

Advanced Area Fill Options

From the Area dialog, you can fine-tune the area fill of an object in greater detail.

Creating Your Own Fill Color

Click on the **Colors** tab of the Area dialog (Figure 21). From here, you can modify existing colors or create your own.

Depending on the color model in use (RGB or CMYK can be selected from the pull-down menu), you can change the individual values of the constituent colors—Red, Green, and Blue or Cyan, Magenta, Yellow, and Black. Click on **Add** to add this color to the color table. Clicking the **Modify** button changes the values of the current color (Blue 8, as shown in the upper rectangle) to those on the screen in the lower rectangle. The **Edit** button allows you to fine-tune the color using a palette with visual feedback.

Further explanation about color palettes can be found in Chapter 8 (Tips and Tricks).

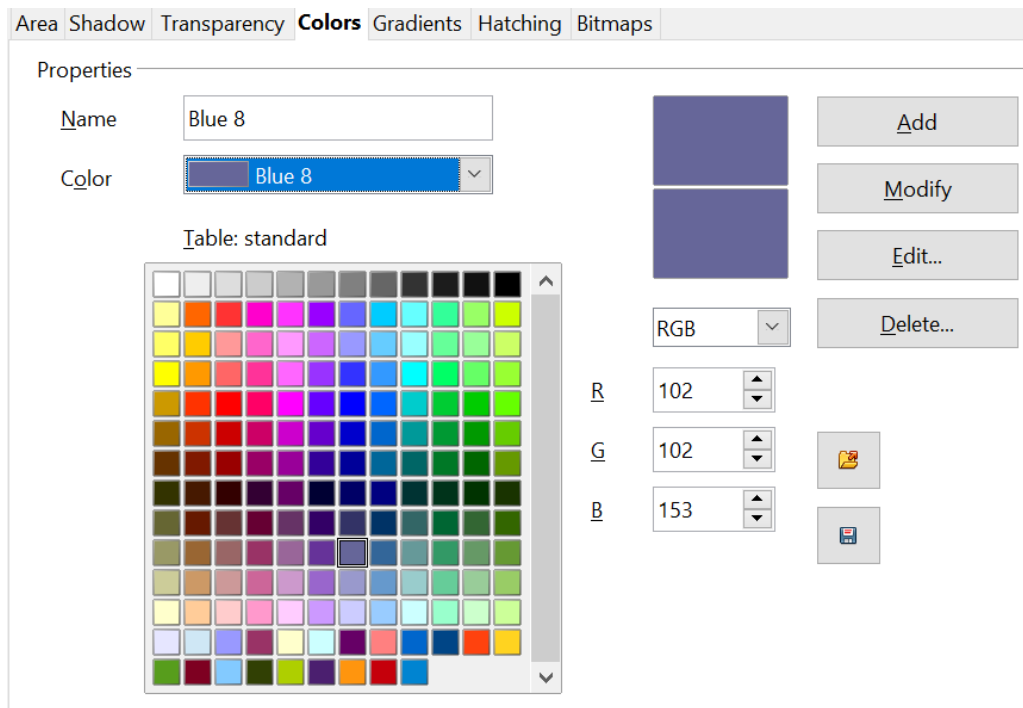


Figure 21: Customizing the color palette

Creating Your Own Gradient

On the Area dialog, click on the **Gradients** tab. From here, you can modify existing gradients or create your own, as we show in Figure 22.

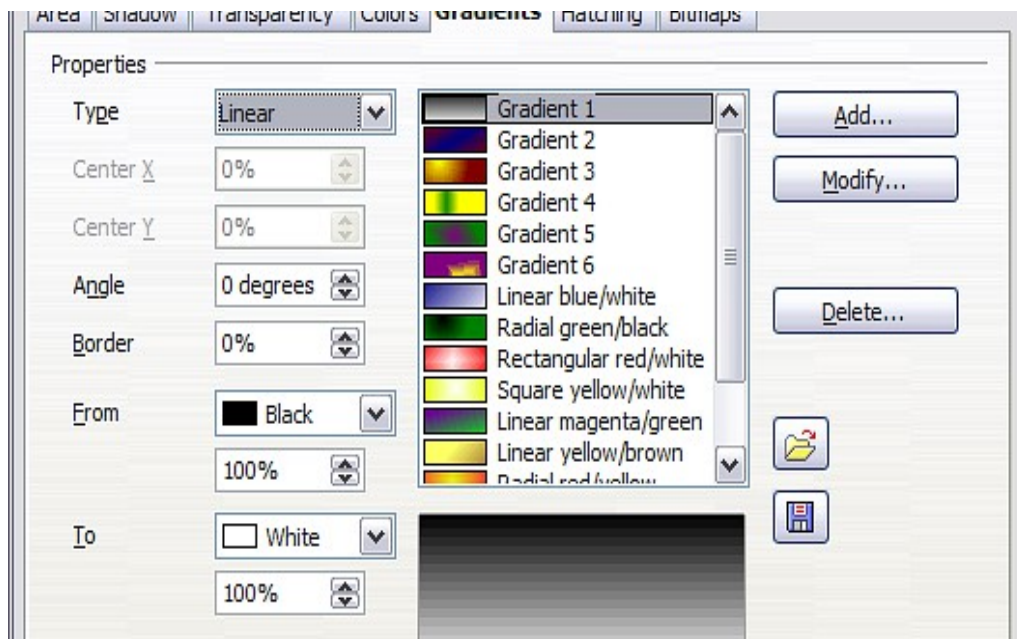


Figure 22: Customizing gradients

A gradient works by creating a smooth transition from one color to another. First, choose two colors (Figure 23).

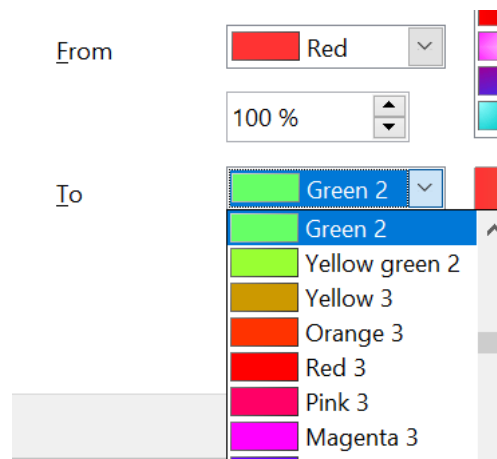


Figure 23: Choosing colors for a gradient

Then choose a type of gradient. There are several available (Linear, Axial, Radial, and so on), and each has up to four different options to specify it in detail. For example, a radial gradient has a center you can specify (Figure 24).



Figure 24: Center option in a radial gradient

Figure 25 shows how an ellipsoid gradient is rotated, moved vertically and horizontally, and the color of the border area is varied.

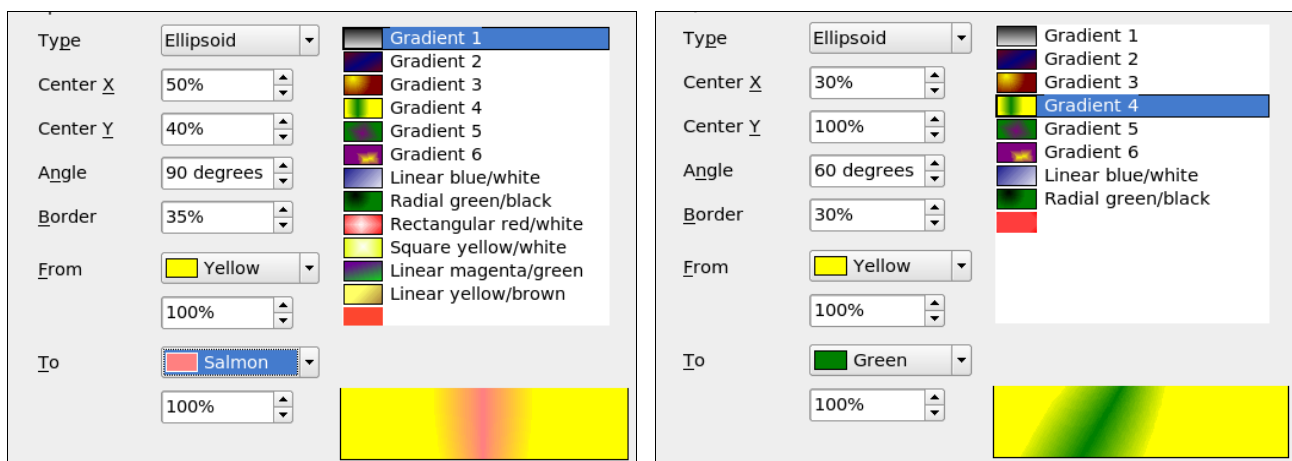


Figure 25: Center, Angle, and Border options for a gradient

Creating Your Own Hatching (Line Pattern)

On the Area dialog, click on the **Hatching** tab (Figure 26). From here, you can modify existing hatchings (line patterns) or create your own.

You can customize options like the spacing between lines, the angle, and the color of the lines. There is no way to edit the line thickness.

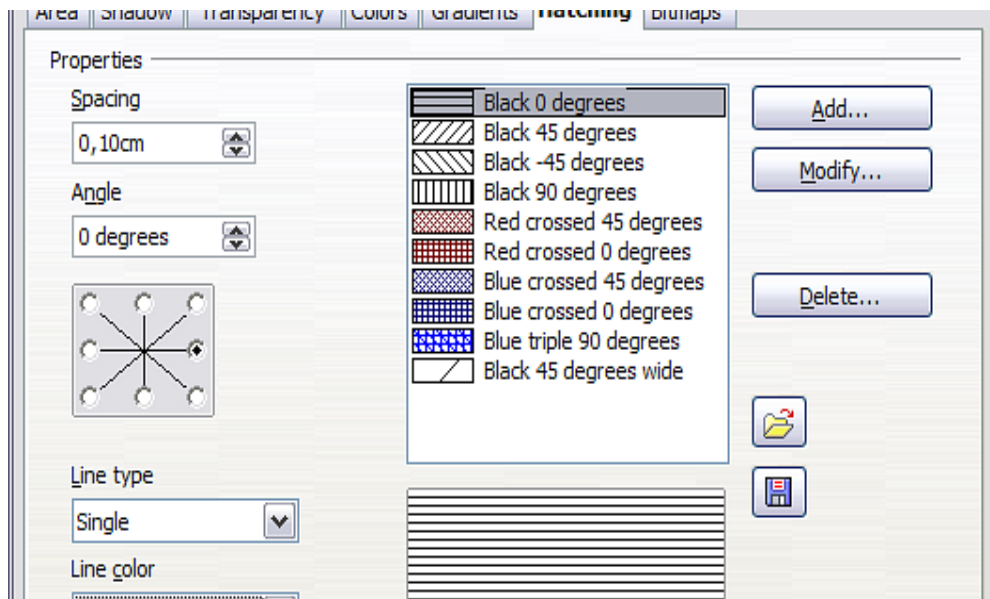


Figure 26: Modifying hatchings

Creating Your Own Bitmap Fill

You can add your own bitmap images to fill an area. First, you need to create the bitmap image. This could be a photo or a sketch you have created in another program. For example, you can draw something with Draw and **export** it as a PNG format image file.

To export a PNG image file: Create the image, select it, then choose **File > Export**, choose **PNG** from the pull-down list of file formats, give the file a name, and save it.

To use an image as a bitmap fill, open the Area dialog and click on the **Bitmaps** tab (Figure 27).

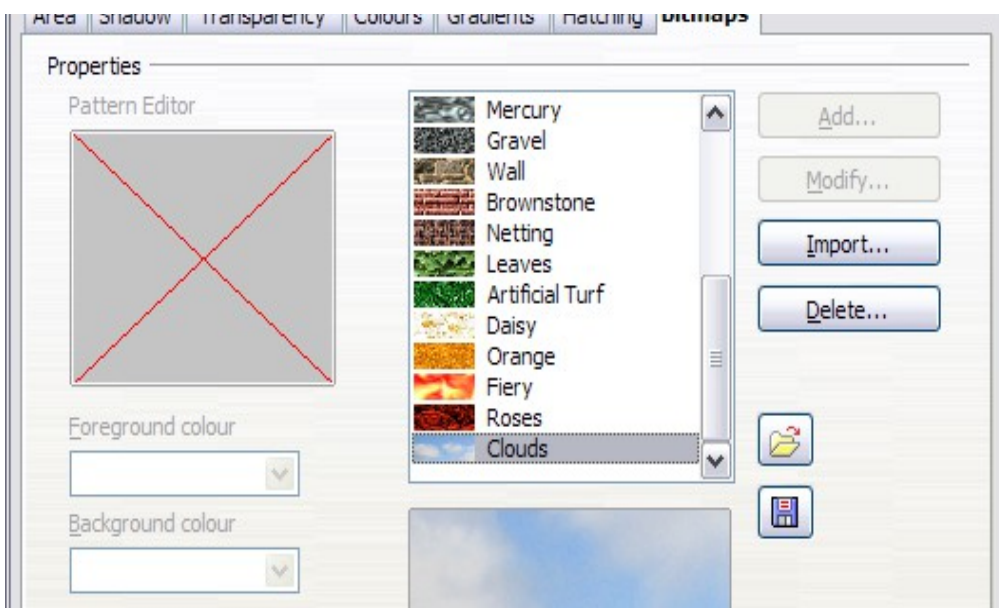


Figure 27: Adding your own bitmap images

From there, you can add new bitmap images to serve as area fills. Click on **Import** and choose a file you previously saved. Give it a name that's easy to remember. The last imported image will appear at the bottom of the pull-down list of bitmap fills. Now you can use that image as an area fill, like Figure 28 shows.

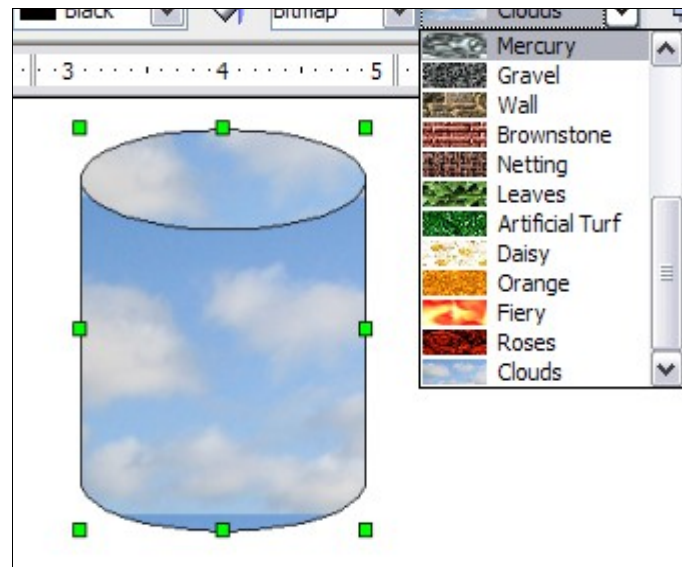


Figure 28: Adding your own bitmap images

Tip

If the imported image appears very small in the preview (Figure 26), you probably forgot to select the image before exporting. In that case, the exported image would be the whole page with a (small) drawing on it.

Customizing Shadows

First, select the object you want to apply a custom shadow to. Open the Area dialog and go to the **Shadow** tab. There, you can customize the shadow's position, distance, and color, like we did in Figure 29.

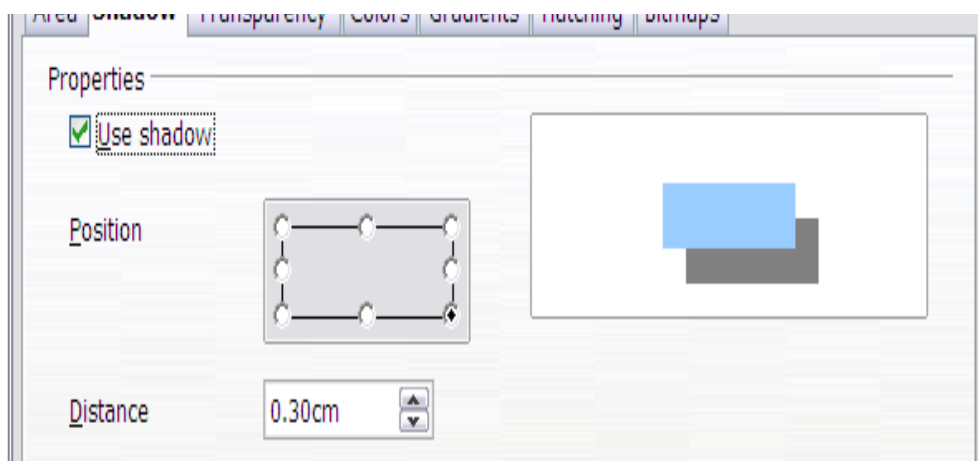


Figure 29: Customizing shadows

Shadows can also have transparency, so the shadow does not completely hide objects behind it; Figure 30 illustrates this effect

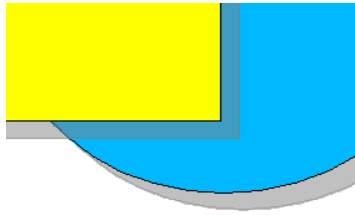
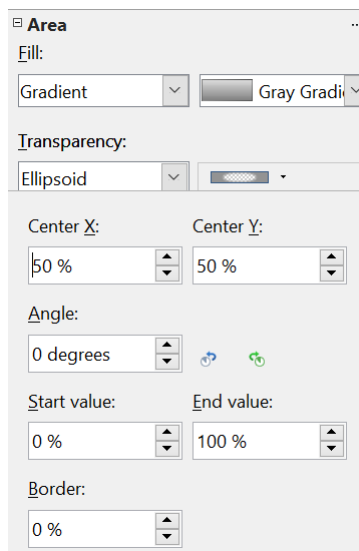


Figure 30: Shadow with 50% transparency

Adding Transparency

You can make objects partly or fully transparent, or even fill them with a varying degree of transparency (as a gradient). This can be done from the Sidebar or from the Area dialog. Figure 31 shows the Area pane of the Sidebar when an object has been selected, the Transparency type has been set to Ellipsoid, and the drop-down list to the right of the Ellipsoid selection has been clicked. Only settings appropriate for the selected transparency type will be displayed. For example, a Solid transparency has only one setting, the fixed transparency value.



*Figure 31:
Transparency settings
on the Sidebar*

On the **Transparency** tab of the Area dialog(Figure 32), choose **Transparency** (for a uniform transparency) or **Gradient** for a gradient transparency.

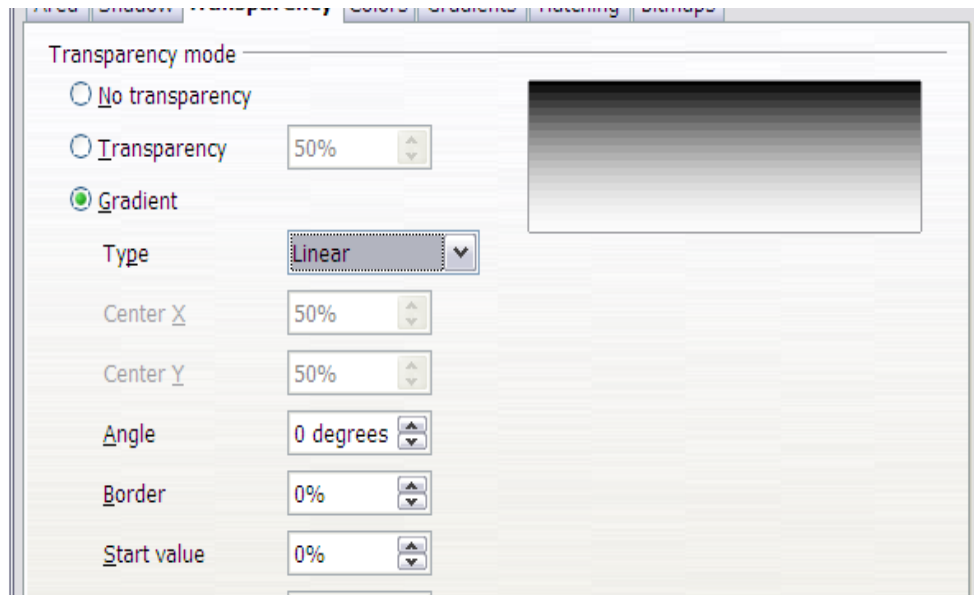


Figure 32: Transparency page of Area dialog

An example of gradient transparency is shown in Figure 33. See also “Dynamic Gradients” on page 26.



Figure 33: Example of gradient transparency

Using Styles

Suppose that you want to apply the same area fill, line thickness, and border to a set of objects. This repetitive process can be greatly simplified by using styles. Styles allow you to define a formatting template and apply it to multiple objects. For more about styles, see Chapter 6 (Introduction to Styles) in the *Writer Guide*.

There are many ways to access the Styles and Formatting tool:

- Click the Styles and Formatting icon on the Sidebar.
- Click on the Styles and Formatting  icon on the Line and Filling toolbar
- Press the *F11* key to open the Styles and Formatting window
- Select **Format > Styles and Formatting**

The tool shown in Figure 34 will be displayed either as a floating window or in the Sidebar. This window can be docked to the left or right side of the main Draw window, if you wish.

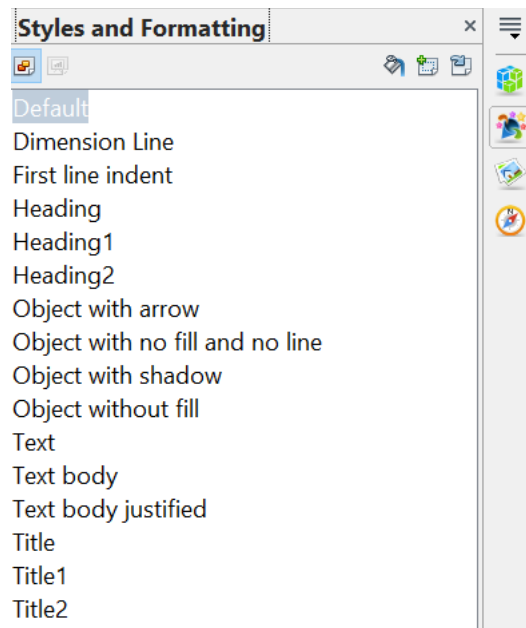


Figure 34: Styles and Formatting window

All of the styles in Draw are Graphics Styles, and they share the same properties. Right-clicking a style and selecting **Modify** opens a dialog with the 14 tabs shown in Figure 35. Each of the supplied styles has a few properties modified from the Default style values, and the style name hints at what those changes are.

Dimensioning				Connector		Alignment			Tabs	
Organizer	Line	Area	Shadowing	Transparency	Font	Font Effect	Indents & Spacing		Text	Text Animation

Figure 35: The tabs of the Graphics Style dialog

If you explore the various tabs, you may feel there is an overwhelming number of settings. Keep in mind that for any particular use, you probably only care about the settings of a very few properties.

Creating a New Style

A new style can be defined from the Graphics Style dialog or from an object. To work with the dialog, right-click in the Styles and Formatting window, either on an empty area or on an existing style. You can then choose **New** or **Modify**. The New option opens the Graphics Styles dialog with the default name Untitled1 and the properties of the style that was clicked, or if no style was chosen, the Default style. You can then set the style name on the Organizer tab and adjust the properties you want. If you select Modify, the process is the same except that the style will be named after the chosen style, and its settings will overwrite the previous properties of that style. In either case, the changes are effective only in the current document.

To create a style from an object, select an object and customize its properties. When you are satisfied, click the **New Style from Selection**  icon in the Styles and Formatting window. This defines a new style based on the selected object. Type a name for the new style and click **OK**.

Applying a Style

Once the new style is defined, you can apply it to other objects. Select another object and double-click on the style name you defined. The new object will acquire the properties of that style.

Tip

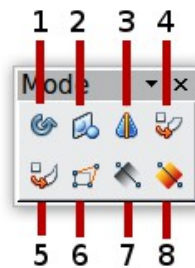
Question: What happens if I modify a style after it has been applied?

Answer: Every object with that style is updated automatically!

Special Effects

First, make sure that the Drawing toolbar is selected (**View > Toolbars > Drawing**).

On the Drawing toolbar, locate the **Effects** icon . Click on the **arrow** next to that icon. This opens a submenu with all the special effect tools (see Figure 36).



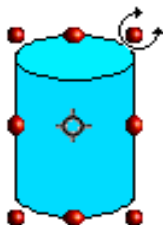
- | | | |
|-------------------------|-------------------------------|----------------|
| 1 Rotate | 4 Set in Circle (perspective) | 7 Transparency |
| 2 Flip | 5 Set to Circle (slant) | 8 Gradient |
| 3 In 3D Rotation Object | 6 Distort | |

Figure 36: Mode tear-off submenu

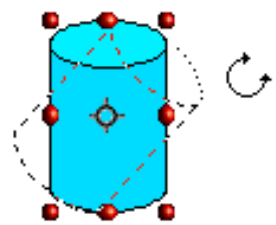
The tools are described in the following sections, except for the 3D rotation tool, which is described in Chapter 7 (Working with 3D Objects).

Rotating an Object

Click on the **Effects**  icon to select the **Rotate** tool. Then select an object. The selected object will have red handles instead of the usual green handles.



Grab one of the handles and move it to rotate the object. The black circle in the middle of the object is the pivot, or center of rotation. You can move the pivot location with the mouse.

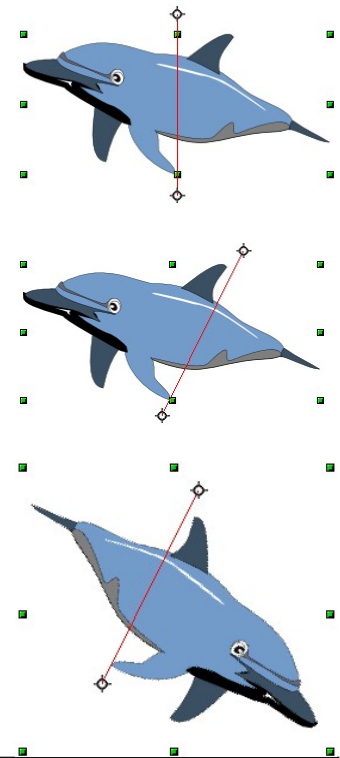


Flip an Object

Select an object and click the **Flip** icon . You will see a dashed line through the middle of the object.

This dashed line is the **axis of symmetry**. The object will be reflected about this line. Move one or both ends of the line with your mouse to set the orientation of the axis.

Then, grab any one of the eight green handles and move it across to the other side of the dashed line. The figure's new position is shown faintly until the mouse is released.



Note If you press the *Shift* key while moving the line, the line will rotate in 45-degree increments.

Mirror Copies

Officially, this useful command does not exist in Draw. It can, however, be easily emulated.

Move the axis of symmetry to the desired location of the mirror axis. Copy the object to the clipboard. Flip the object, then click on an empty area of the Draw screen in order to deselect the object. Paste from the clipboard to put a copy of the object in its original location, and now you have a mirror copy, like that in Figure 37.

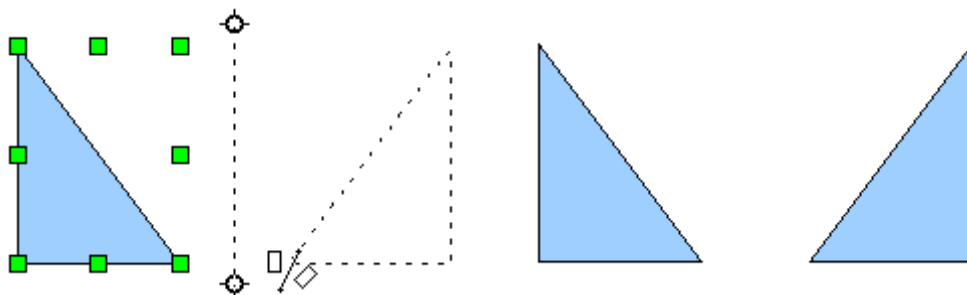


Figure 37: Making a mirror copy of an object

Distorting an Image

Three tools on the **Effects** menu let you drag the corners and edges of an object to distort the image.

The **Distort** tool distorts an object in perspective; the **Set to Circle (slant)** and **Set in Circle (perspective)** tools both create a pseudo three-dimensional effect.

The results of using these tools are shown in the following figures.

Distort an Object

Select an object and click the **Distort** icon . Draw will ask if you want to transform the object into a curve. This is a necessary step before distortion, so click **Yes**. Then you can move the object handles to stretch it.

The corner handles distort, as shown in Figure 38. The midpoint handles distort the figure either horizontally (the handle on the vertical side of the figure) or vertically (the handle on the horizontal side of the figure).

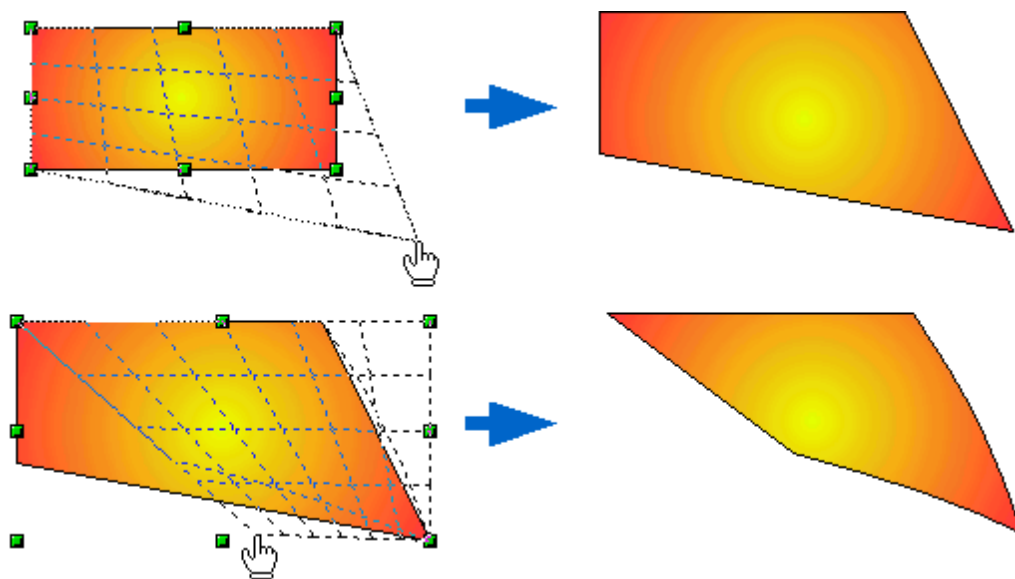


Figure 38: Distort an image

Set in Circle (Perspective)

Select an object and click the **Set in Circle (perspective)**  icon. Draw will ask if you want to transform the object into a curve. This is a necessary step before distortion, so click **Yes**. Then you can move the object handles to create a pseudo-three-dimensional perspective, like that in Figure 39.

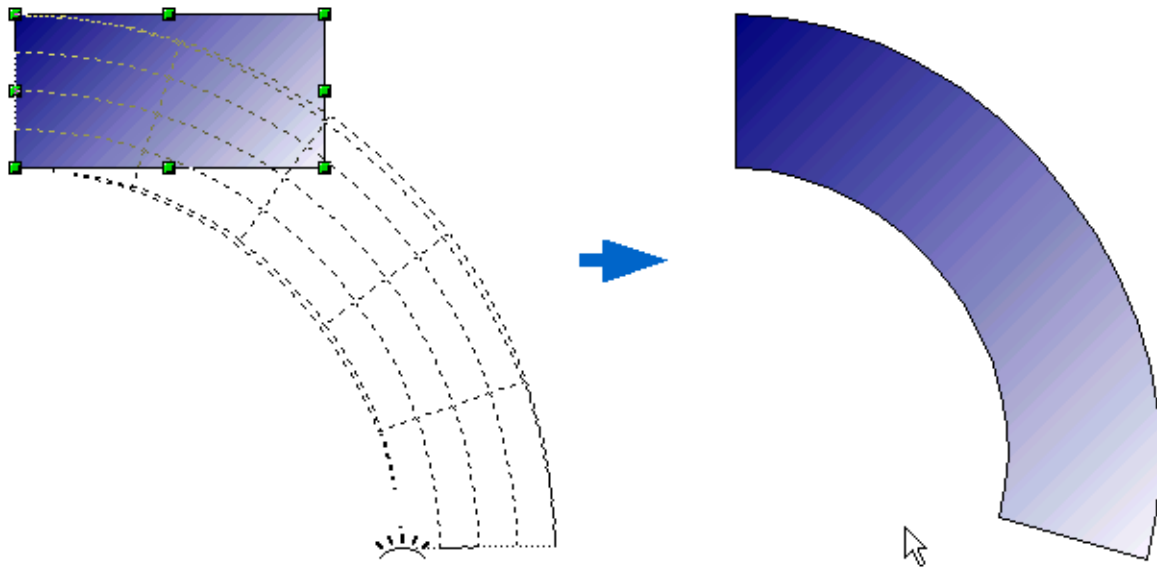


Figure 39: Set an image in a circle with perspective

Set to Circle (Slant)

Select an object and click the **Set to Circle (slant)**  icon. Draw will ask if you want to transform the object into a curve. This is a necessary step before distortion, so click **Yes**. Then you can move the object handles to give a pseudo-three-dimensional slanted perspective, as Figure 40 shows.

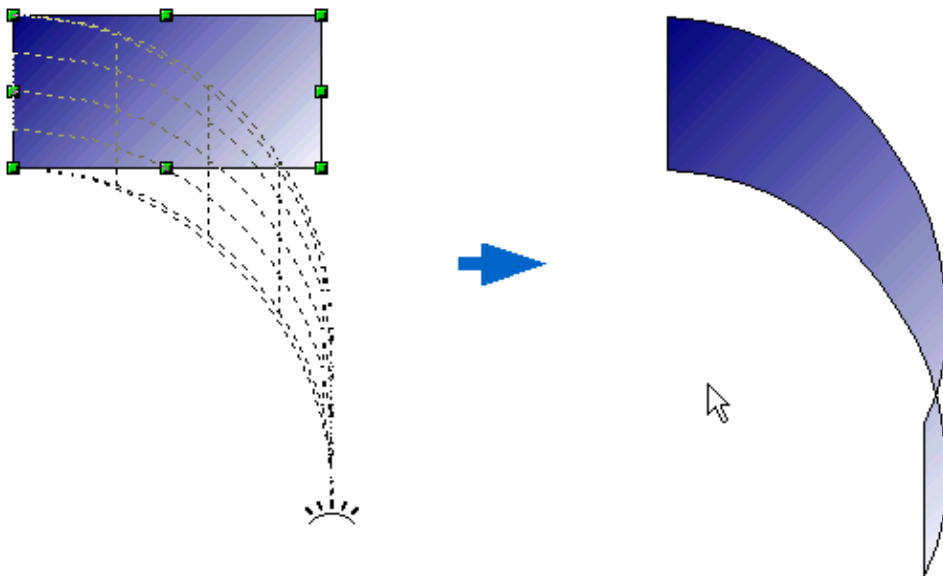


Figure 40: Set an image to a circle with slant perspective

Note

Transforming an object into a curve is a safe operation, but it cannot be reversed other than by clicking the **Undo** button.

Dynamic Gradients

You can control transparency gradients in the same manner as color gradients. Both types of gradient can be used together. With a transparency gradient, the direction and degree of an object's fill color change from opaque to transparent (in a regular gradient, the fill changes from one color to another, but the degree of transparency remains the same).

If you have assigned transparency to an object with a color fill, you can control it by clicking the **Transparency** icon . To define a transparent gradient, select an object, choose a transparency fill from the Sidebar or Line and Filling toolbar. The transparency icon is now active. When you click on this icon, a dashed line connecting two squares appears on the object. Move the two squares to modify the gradient. You can define the gradient direction (vertical, horizontal, or at any angle) and the spot where the transparency begins.

A regular color gradient can be defined in the same manner. Select an object, choose a gradient fill from the Sidebar or Line and Filling toolbar. The **Gradient** icon  is now active. When you click on the gradient icon, a dashed line connecting two squares appears on the object, just as it does for a transparency gradient.

In both cases, click outside the object to set the gradient.

Note	If the transparency and gradient icons are not visible, you can display them using View > Toolbars > Mode. Moving the squares will have different effects, depending on the type of gradient. For example, in a linear gradient, the start and end squares will always be located to either side of the object's center point.
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The three examples in Figure 41 demonstrate how the type and degree of transparency can be controlled.

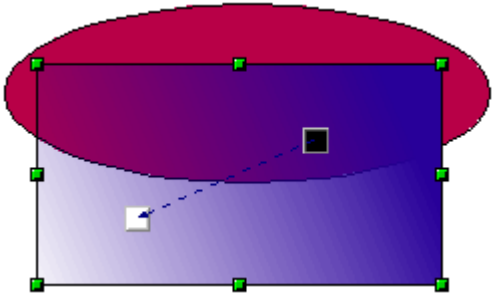
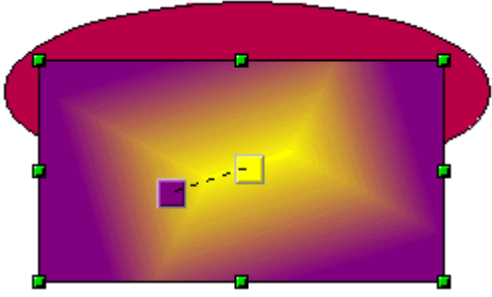
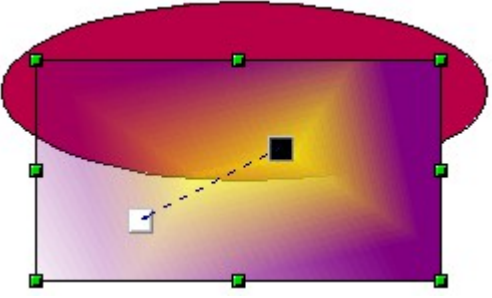
	A single color ellipse and a rectangle with a transparency gradient, covering part of the ellipse. The gradient can be dynamically adjusted, made more transparent by moving the white square or more opaque by moving the black square.
	A rectangle with a color gradient but no transparency, obscuring part of an ellipse. The gradient is adjusted dynamically by moving the squares – the color of the square relates to the increase or decrease in that color.
	A rectangle with both color and transparency gradients, partly covering the underlying ellipse.

Figure 41: Dynamic gradients